

Standing Waves on Strings & in Pipes — Solutions

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Date: 2026-05-06

QUICK RECAP

String / Open pipe	$f_n = \frac{nv}{2L}, \quad \lambda_n = \frac{2L}{n}$	$n = 1, 2, 3, \dots$ — all harmonics.
Closed pipe	$f_n = \frac{nv}{4L}, \quad \lambda_n = \frac{4L}{n}$	$n = 1, 3, 5, \dots$ — odd only.
Universal	$v = f\lambda$	Always check: $v = f_n\lambda_n$.

Solutions

Problem 1

String fixed at both ends. $L = 1.20$ m, $v = 480$ m/s. Find f_1 and λ_1 .

String fixed-fixed

Fundamental

Boundary: nodes at both ends $\Rightarrow f_n = \frac{nv}{2L}, \lambda_n = \frac{2L}{n}$.

$$f_1 = \frac{(1)(480)}{2(1.20)} = \frac{480}{2.40} = 200 \text{ Hz.}$$

$$\lambda_1 = 2L = 2.40 \text{ m.}$$

$$f_1 = 200 \text{ Hz, } \lambda_1 = 2.40 \text{ m}$$

Problem 2

String fixed at both ends, $f_1 = 220$ Hz. List f_2, f_3, f_4 .

Harmonic series

All harmonics allowed: $f_n = n f_1$.

$$f_2 = 2(220) = 440 \text{ Hz.}$$

$$f_3 = 3(220) = 660 \text{ Hz.}$$

$$f_4 = 4(220) = 880 \text{ Hz.}$$

$$f_2 = 440, f_3 = 660, f_4 = 880 \text{ Hz}$$

Problem 3

Open pipe, $L = 0.85$ m, $v = 343$ m/s. Find f_1, f_2 .

Open pipe

All harmonics

Open-open boundary: same formula as fixed string, $f_n = \frac{nv}{2L}$.

$$f_1 = \frac{343}{2(0.85)} = \frac{343}{1.70} \approx 201.8 \text{ Hz.}$$

$$f_2 = 2f_1 \approx 403.5 \text{ Hz.}$$

$$f_1 \approx 202 \text{ Hz, } f_2 \approx 404 \text{ Hz}$$

Problem 4

Closed pipe, $L = 0.85$ m, $v = 343$ m/s. Find f_1 and the next allowed harmonic.

Closed pipe

Odd harmonics

Closed pipe: node at closed end, antinode at open end $\Rightarrow f_n = \frac{nv}{4L}$, with $n = 1, 3, 5, \dots$

$$f_1 = \frac{343}{4(0.85)} = \frac{343}{3.40} \approx 100.9 \text{ Hz.}$$

Next allowed is $n = 3$: $f_3 = 3f_1 \approx 302.6$ Hz.

$$f_1 \approx 101 \text{ Hz, } f_3 \approx 303 \text{ Hz}$$

Problem 5

Same length L . Pipe A open both ends, Pipe B closed one end. Show $f_{1,A} = 2f_{1,B}$ and explain why B has only odd harmonics.

Conceptual

Boundary conditions

Pipe A (open-open): $f_{1,A} = \frac{v}{2L}$.

Pipe B (closed): $f_{1,B} = \frac{v}{4L}$.

Ratio: $\frac{f_{1,A}}{f_{1,B}} = \frac{v/(2L)}{v/(4L)} = \frac{4L}{2L} = 2. \therefore f_{1,A} = 2f_{1,B}. \checkmark$

Why odd-only for closed pipe: the closed end forces a displacement *node* while the open end forces a displacement *antinode*. The shortest distance from a node to the next antinode is $\lambda/4$, so L must equal an *odd* number of quarter-wavelengths: $L = n\lambda/4$ with $n = 1, 3, 5, \dots$. Even n would put a node at the open end, violating the boundary.

Shown: $f_{1,A} = 2f_{1,B}$; closed pipe needs odd $\lambda/4$ count.

Problem 6

Closed pipe, $f_1 = 196$ Hz, $v = 343$ m/s. Find L and λ_1 .

Closed pipe

Solve for L

Closed: $f_1 = \frac{v}{4L} \Rightarrow L = \frac{v}{4f_1}$.

$L = \frac{343}{4(196)} = \frac{343}{784} \approx 0.4375$ m.

$\lambda_1 = 4L \approx 1.75$ m (or directly $\lambda_1 = v/f_1 = 343/196 \approx 1.75$ m). $\checkmark$

$L \approx 0.438$ m, $\lambda_1 \approx 1.75$ m

Problem 7

String fixed both ends, $L = 0.50$ m, $f_4 = 880$ Hz. Find v and λ_4 .

String fixed-fixed

Solve for v

$$f_n = \frac{nv}{2L} \Rightarrow v = \frac{2L f_n}{n}$$

$$v = \frac{2(0.50)(880)}{4} = \frac{880}{4} = 220 \text{ m/s.}$$

$$\lambda_4 = \frac{2L}{4} = \frac{1.00}{4} = 0.250 \text{ m.}$$

$$\text{Check: } v = f_4 \lambda_4 = (880)(0.250) = 220 \text{ m/s. } \checkmark$$

$$v = 220 \text{ m/s, } \lambda_4 = 0.250 \text{ m}$$

Problem 8

Closed pipe, $L = 1.10$ m, $v = 343$ m/s. List the first three allowed harmonic frequencies.

Closed pipe

Odd harmonics

$$f_1 = \frac{v}{4L} = \frac{343}{4(1.10)} = \frac{343}{4.40} \approx 77.95 \text{ Hz.}$$

$$\text{Odd harmonics only: } f_3 = 3f_1 \approx 233.9 \text{ Hz, } f_5 = 5f_1 \approx 389.8 \text{ Hz.}$$

$$f_1 \approx 78.0 \text{ Hz, } f_3 \approx 234 \text{ Hz, } f_5 \approx 390 \text{ Hz}$$

Problem 9

Guitar string, $L = 0.64$ m, $f_1 = 330$ Hz. Find v and λ_2 .

String fixed-fixed

Wave speed

$$f_1 = \frac{v}{2L} \Rightarrow v = 2L f_1.$$

$$v = 2(0.64)(330) = 1.28 \times 330 = 422.4 \text{ m/s.}$$

$$\lambda_2 = \frac{2L}{2} = L = 0.64 \text{ m.}$$

Check: $f_2 = 2f_1 = 660$ Hz, and $v = f_2\lambda_2 = (660)(0.64) = 422.4 \text{ m/s. } \checkmark$

$$v \approx 422 \text{ m/s, } \lambda_2 = 0.640 \text{ m}$$

Problem 10

Pipe resonates at 250 Hz and 750 Hz, but not 500 Hz, $v = 343$ m/s. (a) Open or closed? (b) Find L .

Identify boundary

Pattern recognition

(a) If the pipe were open at both ends, every integer multiple of f_1 would resonate, including 500 Hz. Since 500 Hz is missing, it must be a **closed pipe** (odd harmonics only). Check: $250 = 1 \cdot f_1$, $750 = 3 \cdot f_1 \Rightarrow f_1 = 250$ Hz. The next would be $5f_1 = 1250$ Hz, with 500 and 1000 skipped — consistent.

$$\textbf{(b)} f_1 = \frac{v}{4L} \Rightarrow L = \frac{v}{4f_1} = \frac{343}{4(250)} = \frac{343}{1000} = 0.343 \text{ m.}$$

(a) Closed pipe; (b) $L = 0.343$ m